Experiment-2

Aim: Convert the er-diagram into a relational schema. Write complete mysql Create table statement with primary keys, foreign key, Not Null, Unique, on delete cascade and set null constraints. Insert sample data and demonstrate referencehical integrity.

Source code:

```
Create Database E-commerce;
Use E-commerce;
create Table Customer (
    Customer_ID  int primary key,
    Name varchar(100) not null,
    email varchar(100) unique,
    age int
);
Create Table Seller (
    seller_id int primary key,
    name varchar(100) not null,
    Seller_address varchar(100)
);
Create table Category (
    category_id int primary key,
    category_name varchar(100) Not Null,
    description varchar(100)
);
```

```
Create Table Product (
    Product_id int primary key,
    Product_name varchar(100) not null,
    quantity int,
    price decimal (10,2) not null,
    category_id int
    Foreign key (category_id) References category (category_id)
    on Delete cascade.
);

Create Table Payment (
    payment_id int primary key,
    payment_amount decimal (10,2) not null,
    payment_date Date,
    payment_method varchar(50)
);

Create table Delivery_address (
    address_id int primary key,
    ~~order_date~~ ~~Date~~
    city varchar (50)
    state varchar (50)
    Pincode varchar (10)
);

Create table Orders (
    order_id int primary key,
    order_date Date
    order_amount Decimal (10,2),
    customer_id int,
    seller_id int,
    address_id int
    Foreign key (customer_id)
    References Customer (customer_id)
    On delete cascade;
```

```
Foreign key (seller_id)
  References seller (seller_id)
  Foreign key (payment_id)
     References Payment (payment_id)
   Foreign key (address_id)
     References Delivery_address (address_id)
);
```

Insert into customer values (1, 'Rahul Sharma', 'rahul@gmail.com', 21),
(2, 'Aman Verma', 'aman@gmail.com', 25), (3, 'Priya Singh', 'priya@gmail.com', 22),
(4, 'Neha Gupta', 'neha@gmail.com', 24), (5, 'Rohit Kumar', 'rohit@gmail.com', 25);

Insert into seller values (1, 'Tech World', 'Delhi'), (2, 'Fashion Hub', 'Mumbai'),
(3, 'Home Store', 'Jaipur'), (4, 'Book Point', 'Pune'), (5, 'Gadget Zone', 'Bangalore');

Insert into category values (1, 'Electronics', 'Electronic Products'),
(2, 'Clothing', 'Clothes and fashion'), (3, 'Home', 'Home appliances'),
(4, 'Books', 'Books and study material'), (5, 'Accessories', 'Mobile and computer accesories');

Insert into Product values (101, 'Laptop', 10, 55000.00, 1), (102, 'T-Shirt', 25, 799.00, 2), (103, 'Mixer', 15, 2500.00, 3), (104, 'DBMS Book', 20, 650.00, 4)
(105, 'Keyboard', 30, 1200.00, 5);

Insert into Payment values (201, 55000.00, '2026-09-01', 'UPI')
(202, 799.00, '2026-09-02', 'card'), (203, '2500-00', '2026-09-03', 'Cash')
(204, 650.00, '2026-09-04', 'UPI'), (205, '1200.00', '2026-09-05', 'Net Banking');

Insert into Delivery_address values (301, 'Gurgaon', 'Haryana', '122001'),
(302, 'Mumbai', 'Maharashtra', '400001'), (303, 'Jaipur', 'Rajasthan', 302001),
(304, 'Pune', 'Maharashtra', 411001), (305, 'Bangladesh', 'karnataka', 560001);

Insert into Orders values (401, '2026-09-01', 55 000.00, 1, 1, 201, 301)
(402, '2026-09-02', 799.00, 2, 2, 202, 302)
(403, '2026-09-03', 2500.00, 3, 3, 203, 303)
(404, '2026-09-04', 650.00, 4, 4, 204, 304)
(405, '2026-09-05', 1200.00, 5, 5, 205, 305).

Outcome

1. Customer

customer_id	Name	~~email~~ email	age
1	Rahul Sharma	rahul@gmail.com	21
2	Aman Verma	aman@gmail.com	23
3	Priya Singh	priya@gmail.com	22
4.	Neha gupta	neha@gmail.com	24
5	Rohit Kumar	rohit@gmail.com	25.

2. Seller

Seller_id	name	seller_address
1	Tech World	Delhi
2	Fashion Hub	Mumbai
3.	Home Store	Jaipur
4.	Book Point	Pune
5.	Gadget zone	Bangalore.

3 Category

category-id	category-name	description
1	Electronics	Electronic Product
2	Clothing	clothes and fashion
3	Home	Home appliances
4	Books	Books and study material
5	Accesories	Mobile and computer accesories

4. Product

Product_id	Product_name	quantity	price	category-id
101	Laptop	10	55000.00	1
102	T-Shirt	25	799.00	2
103	Mixer	15	2500.00	3
104	DBMS Book	20	650.00	4
105	Keyboard	30	1200.00	5

5 Payment

Payment-id	Payment_amount	Payment-date	Payment_method
201	55000.00	2026-09-01	UPI
202	799.00	2026-09-02	Card
203	2500.00	2026-09-03	Cash
204	650.00	2026-09-04	UPI
205	1200.00	2026-09-05	Net Banking

6. Delivery_Address

address-id	city	state	pincode
301	Gurgaon	Haryana	122001
302	Mumbai	Maharashtra	400001
303	Jaipur	Rajasthan	302001
304	Pune	Maharashtra	411001
305	Bangalore	Karnataka	560001

7. Orders

order_id	order_date	order_amount	customer_id	seller_id	payment_id	add_id
401	2026-09-01	55000.00	1	1	201	301
402	2026-09-02	799.00	2	2	202	302
403	2026-09-03	2500.00	3	3	203	303
404	2026-09-04	650.00	4	4	204	304
405	2026-09-05	1200.00	5	5	205	305